

Module 06: Learning — Machine Learning and Neural Networks

Learning Objectives

- 1. Distinguish between **supervised**, **unsupervised**, and **reinforcement learning**.
- 2. Explain how a neural network learns by adjusting weights through **backpropagation**.
- 3. Connect machine learning to Active Inference: both are mechanisms for reducing prediction error by updating internal parameters.

Introduction

Machine learning is the technology that lets computers improve at tasks without being explicitly programmed for each case. Instead, the computer learns from *data*. This module introduces the three paradigms of machine learning and shows how they all share a deep connection with Active Inference: learning is updating a model to reduce prediction error.

Key Concepts

1. Three Paradigms of Machine Learning

Paradigm	Input	Goal	Example
Supervised	Data + labels (correct answers)	Predict the label for new data	Email spam filter (label: spam/not spam)
Unsupervised	Data only (no labels)	Find hidden structure (clusters, patterns)	Customer segmentation (group similar shoppers)
Reinforcement	Actions + rewards	Learn a policy that maximizes total reward	Game AI (learn to beat Atari games)

2. Neural Networks and Backpropagation

A **neural network** is a function with millions of adjustable parameters (weights). Training works like this:

1. **Forward pass:** Feed an input through the network, get a prediction
2. **Loss calculation:** Compare the prediction to the correct answer (prediction error!)
3. **Backward pass (backpropagation):** Calculate how much each weight contributed to the error
4. **Update:** Adjust weights using gradient descent ($w \leftarrow w - \eta \cdot \frac{\partial L}{\partial w}$)

This is exactly the Active Inference loop: predict, compare, update. The entire field of deep learning is built on this simple cycle.

3. Overfitting and Regularization

A model that memorizes the training data perfectly but fails on new data is **overfitting**. It is like a student who memorizes answers without understanding the concepts. **Regularization** techniques (dropout, weight decay, early stopping) penalize complexity — forcing the model to learn general patterns. This mirrors the complexity-accuracy trade-off from the Active Inference free energy equation.

Applications

- **Image Recognition:** A CNN trained on ImageNet (14 million labeled images) learns to distinguish 1,000 categories — from goldfish to schooners to laptops. Its layers build increasingly abstract representations.
- **Spotify Recommendations:** A collaborative filtering algorithm learns your music taste by comparing your listening history to millions of other users. The prediction error ("you skipped this song") updates the model.

Discussion Questions

1. If you train a spam filter on only English emails, what happens when it encounters a Spanish email? What type of prediction error is this?
2. Why is overfitting considered "bad learning"? Connect this to the concept of a generative model that is too specific.

Summary

Machine learning trains models by minimizing prediction error on data. Neural networks learn through backpropagation — the computational equivalent of Active Inference's predict-compare-update loop. Regularization prevents overfitting by penalizing complexity.

References

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. Chapters 6-8.